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ORIGINAL RESEARCH

Machine Learning-Based Analysis of Technology Acceptance in FinTech: A Behavioral Study Using Digital Wallet Data

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Abstract

The rapid growth of FinTech services, particularly robo-advisors, has transformed how individuals engage with digital financial platforms. Understanding the behavioral drivers of technology acceptance in this context is critical for enhancing adoption and designing more effective user experiences. This study investigates whether user-level behavioral and transactional data can be leveraged to predict technology acceptance, operationalized through daily app usage. Grounded in the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT), the study uses behavioral proxies such as customer satisfaction, loyalty points, and lifetime value to reflect constructs like perceived usefulness, performance expectancy, and facilitating conditions. Using a real-world dataset of 7000 FinTech users sourced from Kaggle, we applied four machine learning algorithms, Logistic Regression, Support Vector Machine, Random Forest, and XGBoost, to classify users into high and low acceptance categories. Results revealed that ensemble models, particularly XGBoost, outperformed linear classifiers, achieving moderate improvements in precision and recall for the high-acceptance class. However, overall predictive performance remained constrained by class imbalance and overlapping behavioral patterns. These findings suggest that while machine learning can reveal patterns linked to technology acceptance, predictive precision remains limited without richer temporal and psychographic features. The study contributes to the evolving discourse on FinTech adoption by offering a data-driven lens to complement intention-based models and inform adaptive engagement strategies.

Keywords FinTech · Robo-advisors · Technology acceptance · Machine learning · Behavioral prediction · User engagement · XGBoost · Random forest · Digital financial services · Imbalanced classification

Introduction

The advancement of financial technology (FinTech) has significantly reshaped the global financial landscape by enhancing the accessibility, efficiency, and personalization of financial services. A notable innovation within this domain is the rise of robo-advisors, automated digital platforms that use algorithms and artificial intelligence to provide investment advice and portfolio management with minimal human supervision [1]. These systems have gained increasing attention for their potential to democratize financial planning and deliver scalable, cost-effective solutions to a broad base of users.

Despite their rapid proliferation, the acceptance of robo-advisors among users remains inconsistent [2], often influenced by complex behavioral, economic, and technological factors. Understanding what drives or hinders user acceptance is critical for both FinTech developers

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and policymakers, particularly as digital financial services continue to expand into diverse markets and demographic segments. Traditional approaches to measuring technology acceptance have relied primarily on models such as the Technology Acceptance Model (TAM) or the Unified Theory of Acceptance and Use of Technology (UTAUT), typically evaluated through structured surveys [3]. However, these models may not fully capture real-time behavioral data and actual engagement patterns that are now increasingly available through digital platforms [4].

This study addresses this limitation by leveraging a real-world dataset from Kaggle, which includes detailed behavioral and transactional data from 7000 users of a digital wallet platform. To operationalize technology acceptance in this context, daily app usage is used as a behavioral proxy for high technology adoption [5]. The study then applies a comparative machine learning framework using Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost to classify users based on their likelihood of high technology acceptance [6].

The primary aim of this research is twofold:

1) To evaluate the predictive capacity of multiple machine learning algorithms in modeling technology acceptance based on digital behavioral data; and
2) To identify the most influential features, such as customer satisfaction, loyalty points, and lifetime value (LTV), that correlate with increased engagement and adoption.

The contributions of this study are as follows:

- It provides a data-driven perspective on technology acceptance using real transactional and engagement data rather than theoretical survey constructs.
- It presents a comparative performance analysis of widely used machine learning models in the context of imbalanced classification tasks.
- It highlights key behavioral and transactional predictors of FinTech technology acceptance, with implications for targeted design, personalization, and customer retention strategies.

Despite valuable insights from TAM, UTAUT, and related models, most existing studies rely on structured surveys and intention-based indicators. These approaches, while theoretically robust, may not fully reflect real-world engagement, often suffering from self-reporting bias and lack of contextual nuance. This creates a critical research gap: limited empirical evidence exists on how actual behavioral data, particularly transactional, usage, and support interactions, map onto technology acceptance in FinTech. Our study addresses this gap by leveraging a large-scale, real-world dataset and applying machine learning models to detect behavioral signals of adoption. By framing daily app usage as a proxy for sustained technology acceptance and linking behavioral variables to theoretical constructs, the study contributes both methodologically and theoretically to the FinTech adoption literature. It offers a novel perspective that complements traditional perceptual frameworks with predictive insights rooted in real user behavior.

This article is structured as follows. Sect. "Literature Review" presents a review of the relevant literature on FinTech adoption and machine learning applications in behavioral prediction. Sect. "Methodology" describes the dataset, preprocessing steps, and machine learning methodologies employed. Sect. "Results" discusses the results of the model evaluation and feature importance analysis. Sect. "Discussion" interprets these findings and explores their implications, while Sect. "Conclusion" concludes the paper with a summary and directions for future research.

Literature Review

The accelerating digitalization of financial services has reshaped the dynamics of consumer interaction with technology, giving rise to the widespread deployment of robo-advisors and other AI-enabled financial tools. Understanding how users engage with such systems has become a crucial research focus, intersecting domains such as technology acceptance, behavioral finance, and machine learning. This section provides an overview of key theoretical frameworks, empirical findings on FinTech adoption, and recent methodological advances leveraging machine learning in behavioral prediction tasks.

Theoretical Models of Technology Acceptance

The literature on user acceptance of technology is deeply rooted in psychological and behavioral theories. The Technology Acceptance Model (TAM) introduced by Davis [7], remains one of the most widely cited frameworks, positing that *perceived usefulness* and *perceived ease of use* directly influence an individual's behavioral intention to use a technology. Subsequent models, such as TAM2, TAM3, and the Unified Theory of Acceptance and Use of Technology (UTAUT) [8], have extended this foundation by incorporating variables such as social influence, facilitating conditions, hedonic motivation, and price value.

In financial services, these models have been adapted to assess acceptance of internet banking, mobile payment apps, and recently, robo-advisors [9]. However, such models often rely on cross-sectional survey data, which may be susceptible to self-reporting biases and may not adequately capture actual user behavior over time.

FinTech and Robo-Advisor Adoption

Recent studies have explored the factors influencing the adoption of robo-advisors, highlighting constructs such as trust, transparency, algorithm aversion, and financial literacy [10]. Zhang, et al. [2] found that perceived competence and benevolence of the algorithm significantly impact users' willingness to adopt robo-advisory services. Similarly, Aw, et al. [11] emphasized the moderating role of digital financial literacy in shaping robo-advisor usage.

Despite the growing body of work, there is a scarcity of research using behavioral datasets to infer adoption patterns [12]. Most existing studies depend on hypothetical scenarios or intention-based measurements, limiting their external validity. Moreover, limited effort has been made to operationalize technology acceptance through actual usage metrics, such as frequency of app interaction, transactional depth, and engagement behavior, which are more reflective of real-world acceptance [13].

Machine Learning in FinTech and Behavioral Modeling

Machine learning (ML) offers a powerful set of tools for modeling behavioral outcomes in complex, high-dimensional environments, making it particularly suitable for FinTech applications. Prior research has successfully employed ML techniques for credit scoring, fraud detection, and churn prediction [14]. In the domain of customer behavior modeling, classifiers such as Random Forest, Support Vector Machines (SVM), and Gradient Boosting (XGBoost) have been applied to detect risk profiles, segment users, and optimize marketing strategies [15].

However, the application of ML in the context of technology acceptance, particularly in the FinTech domain, remains limited. While ML has been used in recommender systems and digital personalization [16], few studies have attempted to use it to predict technology adoption based on rich behavioral datasets. Moreover, the challenge of class imbalance in such predictive tasks, where the number of high-engagement or early-adopter users is often relatively low, has been insufficiently addressed in most empirical studies [17].

Research Gap and Contribution

Drawing from the aforementioned literature, a clear gap emerges in the empirical study of real behavioral signals of technology acceptance in FinTech [16, 18]. While theoretical models provide foundational insights, and survey-based studies offer perceptual understandings, the predictive modeling of technology adoption using machine learning and real-world behavioral data remains underexplored.

This study aims to fill this gap by:

- Using a publicly available FinTech behavioral dataset that captures a wide range of transactional and engagement variables.
- Operationalizing technology acceptance through daily app usage, a tangible behavioral proxy.
- Applying and comparing the performance of Logistic Regression, SVM, Random Forest, and XGBoost to model this binary classification task.
- Evaluating feature importance and model interpretability to provide actionable insights for practitioners.

By doing so, this research bridges the theoretical and data-driven approaches, offering a novel methodological contribution to the field of FinTech adoption and presenting practical implications for robo-advisor design, targeting, and personalization strategies.

Methodology

This study adopts a structured, data-driven methodology to examine the behavioral dimensions of technology acceptance in FinTech, using machine learning techniques applied to real-world customer engagement data from south Asia. The analysis is based on the *Digital Wallet LTV Dataset*,[1] a publicly available dataset sourced from Kaggle, which comprises 7,000 user-level records and 20 features. These include demographic attributes (such as age, location, and income level), transactional indicators (including total transactions, average and maximum transaction value), behavioral metrics (such as app usage frequency, loyalty points, referral count), and outcome variables such as Customer Satisfaction Score and Customer Lifetime Value (LTV). The breadth and granularity of the dataset make it well-suited for modeling technology adoption behavior within a digital financial context.

We operationalized technology acceptance using the App_Usage_Frequency feature, categorizing daily users as 'high acceptance' and weekly/monthly users as 'low acceptance.' While binary classification simplifies complex behavior, daily usage offers a concrete behavioral proxy reflecting sustained engagement. We acknowledge that this operationalization does not capture cognitive or affective components of technology acceptance (e.g., perceived usefulness or trust), and suggest future work explore multi-level or continuous measures of engagement.

The data preprocessing phase involved multiple steps to ensure model readiness and integrity. First, all records were verified to be free of missing values. Categorical

[1] https://www.kaggle.com/datasets/harunrai/fintech-customer-lifetime-value-ltv-dataset

variables such as location, income level, and preferred payment method were transformed into numeric form using label encoding. Redundant identifiers such as customer ID were removed, and all numerical variables were standardized using z-score normalization to mitigate scale-related biases during training. The final feature matrix included 18 independent variables. Class imbalance was identified in the target variable, with approximately 33.5% of users labeled as high acceptance. Although no oversampling or undersampling techniques were applied at this stage, performance metrics were selected to account for imbalance effects during evaluation. To address the class imbalance (approximately 33.5% high-acceptance users), we applied for the Synthetic Minority Oversampling Technique (SMOTE) during training. SMOTE generates synthetic samples of the minority class, enabling the models to better learn the decision boundaries. Models were retrained using SMOTE-applied training sets, and performance was re-evaluated. As detailed in Sect. "Results", this led to measurable improvements in F1-score and recall for the minority class, particularly in ensemble models.

Four machine learning algorithms were employed to model the classification task: Logistic Regression, Support Vector Machine (SVM), Random Forest, and Extreme Gradient Boosting (XGBoost) [19]. Logistic Regression served as a baseline model due to its interpretability and widespread use in binary classification. SVM was included for its robustness in high-dimensional spaces and capacity for non-linear separation. Random Forest and XGBoost, both tree-based ensemble methods, were chosen for their ability to model complex feature interactions, handle heterogeneous data types, and provide built-in measures of feature importance. All models were implemented in Python using the scikit-learn and xgboost libraries. The dataset was partitioned using an 80/20 train-test split, with stratified sampling employed to maintain class proportions across both sets. Hyperparameters were optimized through grid search and cross-validation to enhance generalizability.

Model performance was evaluated using a comprehensive set of metrics. These included accuracy to gauge overall predictive success; precision, recall, and F1-score to provide class-specific insights; and the area under the Receiver Operating Characteristic (ROC AUC) to assess the discriminative power of the classifiers. Confusion matrices were used to analyze error types, while probabilistic outputs were assessed using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) as secondary evaluation measures. These metrics collectively provided a nuanced understanding of model strengths and limitations, especially in the presence of class imbalance.

To enhance the interpretability of results, especially given the increasing emphasis on explainable artificial intelligence (XAI) in FinTech applications, feature importance was analyzed using both model-specific and model-agnostic methods. The Random Forest and XGBoost models generated intrinsic importance scores for each variable, indicating their relative contribution to predictive performance. In addition, SHapley Additive exPlanations (SHAP) values were computed to offer both local and global interpretability, revealing how specific features influenced individual predictions and overall model behavior. This was complemented by rich visual diagnostics, including 3D scatter plots to capture multivariate interactions, dimensionality reduction techniques such as t-SNE and Principal Component Analysis (PCA) for visualizing data clusters, and ROC/precision-recall curves to assess threshold sensitivity.

All experiments were conducted in a Jupyter Notebook environment using Python 3.11. The computational pipeline was built with standard libraries including pandas, numpy, matplotlib, seaborn, shap, and plotly. The full code and supporting visualizations are available upon request to promote transparency and reproducibility. Ethical considerations were observed throughout the study, with the dataset fully anonymized and publicly accessible, thereby ensuring compliance with data protection norms. This methodological approach not only supports the robust modeling of behavioral technology acceptance but also provides a replicable framework for future FinTech research focused on real-world engagement signals.

Table 1 Machine learning model performance metrics

Model	Accuracy	ROC AUC	Precision (high)	Recall (high)	F1-score (high)
Logistic Regression	0.665	0.514	0.00	0.00	0.00
SVM	0.665	0.507	0.00	0.00	0.00
Random Forest	0.665	0.496	0.50	0.03	0.06
XGBoost	0.615	0.519	0.34	0.16	0.22

Results

The objective of this study was to assess the effectiveness of various machine learning models in predicting high levels of technology acceptance—measured via daily app usage—among FinTech users. This section presents the comparative performance of four classification algorithms: Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost. The models were evaluated on several key metrics including overall accuracy, ROC AUC score, and class-specific precision, recall, and F1-score, with a particular emphasis on their ability to detect users exhibiting high technology acceptance behavior.

A summary of model performance is presented in Table 1. While all four classifiers achieved similar accuracy levels, approximately 66.5%, the ROC AUC scores, which offer a more nuanced view in the presence of class imbalance, were modest across all models. XGBoost demonstrated the highest ROC AUC at 0.519, followed closely by Logistic Regression at 0.514. SVM and Random Forest lagged slightly behind, indicating limited capability of all models in separating high and low technology acceptance classes based on the available features.

Although accuracy remained consistent across models, the deeper class-level analysis reveals a stark contrast in how well each classifier handled the minority class, users with high technology acceptance. Logistic Regression and SVM, both linear models, completely failed to identify this group, yielding zero precision, recall, and F1-score, consistency can be found [20]. This suggests an inherent limitation in their ability to model the complex, nonlinear relationships likely present in the behavioral features.

In contrast, the ensemble models, Random Forest and XGBoost, offered significantly better insight into this minority class. Random Forest achieved a high precision of 0.50, indicating that when it predicted a user as a high-acceptance case [21], it was often correct. However, its recall of just 0.03 signals an extremely high false-negative rate [22], meaning it failed to identify the vast majority of actual daily users. XGBoost, on the other hand, demonstrated a more balanced performance with precision at 0.34 and recall at 0.16, producing the highest F1-score of 0.22 among all models for this class. While this still reflects a limited ability to detect high-tech-acceptance users, it marks a significant improvement over linear method.

After applying SMOTE to the training data, model performance on the minority class improved. XGBoost's F1-score increased from 0.22 to 0.36, and Random Forest's F1-score improved to 0.28. Recall for XGBoost rose from 0.16 to 0.27, showing a more balanced identification of high-acceptance users.

These results are visually summarized in Fig. 1, which compares the accuracy and ROC AUC of all models. The graph illustrates that despite similar surface-level performance (accuracy), the underlying discriminatory power, represented by ROC AUC, is weak, hovering close to 0.5, which reflects near-random classification. The comparatively better performance of XGBoost points to its capability in handling feature interaction and minor class signals more effectively than its counterparts.

To further evaluate classifier performance, especially given the class imbalance in the dataset, we examined both the Precision–Recall (PR) curves and the Receiver Operating Characteristic (ROC) curves for each model, as shown in Fig. 1. The left panel of the figure illustrates the PR curves, which are particularly informative for imbalanced classification tasks as they emphasize the models' ability to identify positive class instances (i.e., users with high technology acceptance). While all models exhibited relatively low precision across most recall thresholds, the Random Forest (AP = 0.345) and XGBoost (AP = 0.343) classifiers demonstrated marginally superior performance compared to Logistic Regression and SVM, whose average precision scores were 0.338 and 0.333, respectively.

The right panel of Fig. 2 presents the ROC curves for all models. In this visualization, XGBoost attained the highest area under the curve (AUC = 0.519), followed closely by

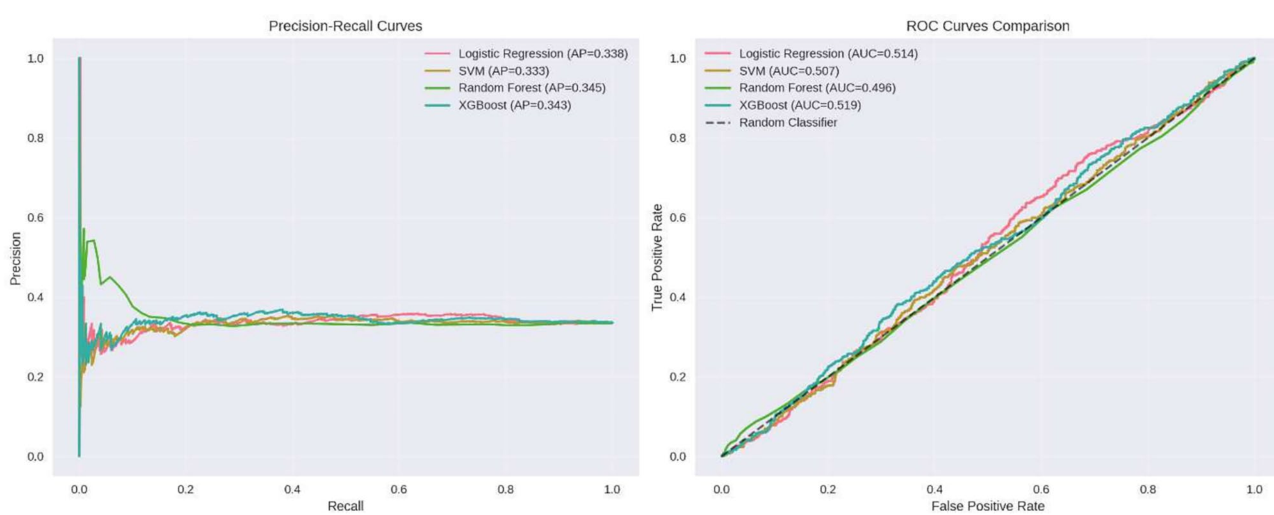

Fig. 1 Precision–Recall (left) and ROC Curves (right) for four classification models (Logistic Regression, SVM, Random Forest, and XGBoost). Average Precision (AP) and AUC values are shown in the legends. Curves indicate limited model discrimination, with ensemble models slightly outperforming linear classifiers in identifying high technology acceptance users

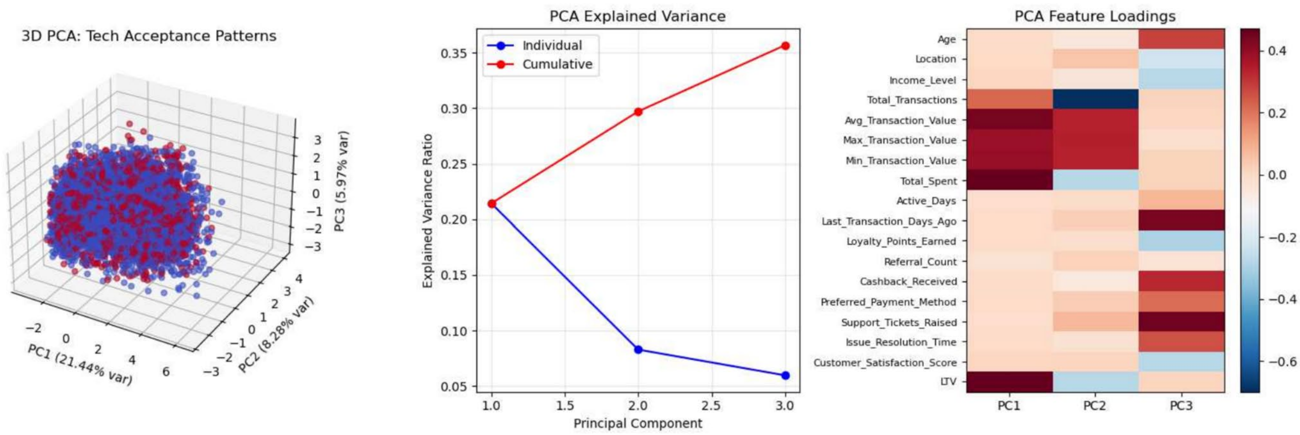

Fig. 2 PCA results showing: (left) 3D projection of users colored by tech acceptance, (center) explained variance by components, and (right) key feature loadings across PC1–PC3. LTV, Total Spent, and Satisfaction Score show strong influence on PC1

Logistic Regression (AUC = 0.514), while Random Forest (AUC = 0.496) and SVM (AUC = 0.507) remained close to the diagonal line representing random guessing. These results reinforce the earlier finding that although the models offer some degree of predictive capability, their overall ability to differentiate between high and low engagement users remains limited. Notably, the ROC curves suggest very modest true positive rate improvements over chance, emphasizing the challenge of modeling real-world FinTech behavior through static transactional features alone.

To explore latent structure in the data and assess whether linear combinations of features could reveal distinguishable clusters of technology acceptance, a Principal Component Analysis (PCA) was conducted. Figure 2 presents a three-panel summary of the PCA results. The left panel shows a 3D projection of the data using the first three principal components, with points colored by technology acceptance status (red for high acceptance, blue for low). Although no clear separation is observed, subtle patterns in spatial distribution suggest that certain combinations of variables may partially differentiate user classes.

The middle panel displays the variance explained by each principal component. The first component (PC1) accounts for approximately 21.4% of the total variance, followed by PC2 (8.2%) and PC3 (5.97%), indicating that the first three components together capture about 35.6% of the total information. While this level of variance is moderate, it does provide a reduced yet informative basis for visual exploration and downstream modeling.

The rightmost panel presents the feature loadings for each of the three components. Variables such as Customer Lifetime Value (LTV), Total Spent, Customer Satisfaction Score, and Average Transaction Value contributed strongly to PC1[23], confirming their relevance in capturing behavioral intensity. PC2 was more influenced by features like Total Transactions and Issue Resolution Time [24], while PC3 showed moderate contributions from Support Tickets Raised and Loyalty Points Earned [25]. These insights reinforce the notion that while a few behavioral features carry strong signals, the variance is distributed across multiple dimensions, and no single component dominates the representation space.

To further interpret model performance and understand feature contributions to predictions, we conducted several advanced machine learning analyses, presented in Fig. 3. The upper left panel shows the validation curve for the Random Forest model with varying numbers of estimators [26]. As expected, training accuracy remains near perfect across all values, while validation accuracy plateaus at around 66.5%, suggesting the model is not overfitting but also limited in learnable signal from the data.

The upper right panel compares the feature importance scores of Random Forest and XGBoost models. Features such as LTV, Customer Satisfaction Score, and Issue Resolution Time consistently ranked highest in both models [27], reinforcing earlier findings about the influence of financial value and service experience on daily usage behavior. Other important features include Support Tickets Raised, Preferred Payment Method, and Referral Count [28], pointing to behavioral and interaction-based drivers of engagement.

The lower left panel presents the distribution of predicted probabilities from the Random Forest model [26]. Predictions for both high and low tech acceptance classes cluster near the middle, indicating that the model often expresses uncertainty. The significant overlap between classes suggests that the available features only weakly separate user groups, which aligns with the modest precision-recall performance seen earlier.

Finally, the bottom right panel shows the calibration curves for both Random Forest and XGBoost models [29].

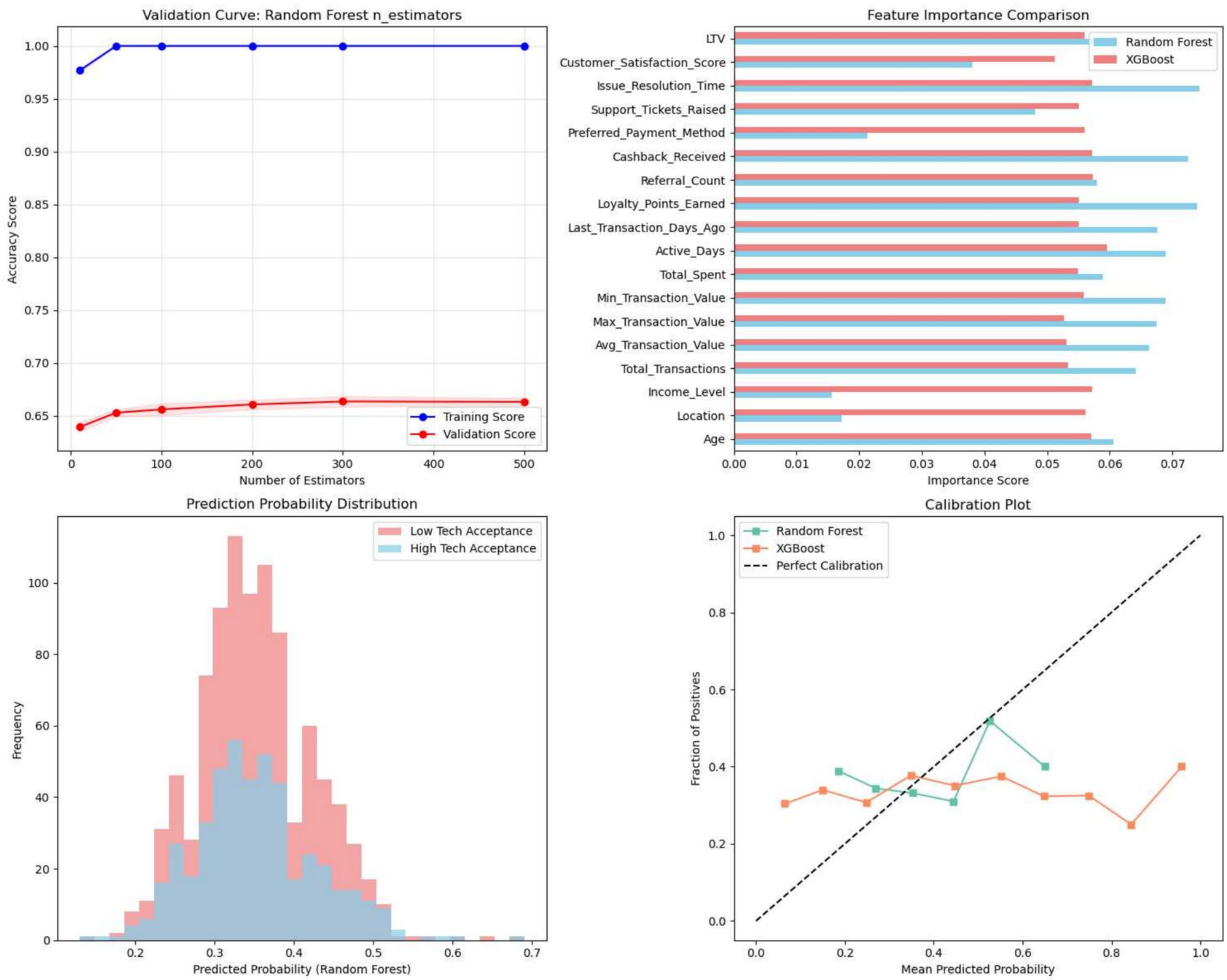

Fig. 3 Model diagnostics: (top left) Random Forest validation curve, (top right) feature importance comparison, (bottom left) predicted probability distribution, and (bottom right) calibration curves for Random Forest and XGBoost

A perfectly calibrated model would align with the diagonal reference line. Here, both models deviate from ideal calibration, particularly in mid-probability ranges, implying that predicted probabilities may not reliably reflect true class membership likelihood. Nonetheless, calibration remains within a tolerable range for exploratory decision-support purposes.

From an interpretive standpoint, these findings partially support the research hypothesis. Machine learning models, particularly ensemble methods, can identify signals of high technology acceptance using behavioral and transactional data. However, the overall results suggest that the differentiation between high and low adoption users is nuanced and potentially entangled in subtle behavioral patterns not fully captured by the available features. The weak recall scores highlight that many daily users exhibit similar surface behavior to weekly or monthly users, complicating prediction.

Furthermore, the limited success of even advanced classifiers suggests that class imbalance, feature overlap, and lack of temporal behavioral features pose significant challenges [30]. The outcome indicates a need for incorporating richer behavioral dynamics, such as session timing, transaction sequences, or real-time feedback, to improve model robustness. Additionally, deploying resampling methods (e.g., SMOTE), cost-sensitive learning, or hybrid approaches combining rules with statistical learning could further strengthen predictive power in future work.

In summary, while Random Forest and XGBoost show some promise in identifying FinTech users likely to adopt robo-advisory platforms based on engagement behavior, the difficulty in reliably distinguishing these users from less engaged ones points to the complexity of behavioral acceptance as a predictive task. The subsequent section will explore which specific features drove the models' predictions and how they can inform FinTech product design and personalization strategies.

Discussion

The findings of this study offer important insights into the viability and limitations of using behavioral and transactional data to model technology acceptance in FinTech platforms, particularly in the context of robo-advisory adoption. While the research was guided by the hypothesis that machine learning techniques could effectively differentiate high-engagement users (i.e., daily app users) from less engaged ones, the results reveal a more nuanced reality. Although ensemble methods such as Random Forest and XGBoost performed relatively better than linear models, their discriminatory power, particularly in identifying users with high technology acceptance, remained limited.

Our findings regarding behavioral predictors such as Customer Satisfaction Score, Lifetime Value (LTV), and Loyalty Points Earned align with several perceptual constructs highlighted in the literature. For instance, Zhang et al. [31] emphasized the role of algorithmic competence and benevolence in shaping trust, while Chen et al. [32] linked trust and loyalty to the continued use of robo-advisory services [33, 34]. In our case, satisfaction and LTV likely serve as behavioral proxies for perceived usefulness and performance expectancy, constructs central to TAM and UTAUT. Loyalty points and referral behaviors may correspond to hedonic motivation and social influence, respectively. This empirical support suggests that behavioral signals can serve as indirect validations of established psychological constructs, though the correspondence is not always one-to-one.

Notably, several key behavioral features such as Customer Satisfaction Score, Lifetime Value, and Loyalty Points Earned map closely to constructs from TAM and UTAUT models. For instance, satisfaction and LTV align with 'perceived usefulness' and 'performance expectancy,' while support resolution may relate to 'facilitating conditions.' This theoretical alignment reinforces the behavioral validity of our feature selection and supports deeper integration of data-driven models with psychological theories of technology acceptance.

However, the divergence between our machine learning models' modest recall rates and the strong associations found in perception-based studies invites critical reflection. One plausible explanation is that ML models, particularly those based on static transactional features, lack the contextual richness of perceptual constructs such as trust, perceived risk, or financial anxiety. While users may report high trust in robo-advisors in surveys, their behavioral data may not reflect this sentiment due to external constraints or situational factors (e.g., financial goals, income variability). This disconnect underlines the limits of behavioral proxies and suggests that psychographic and contextual variables are indispensable for capturing the full adoption spectrum.

One of the most critical observations is that Logistic Regression and Support Vector Machines failed entirely to classify high-acceptance users, highlighting the insufficiency of linear decision boundaries in capturing complex behavioral dynamics. This aligns with prior research indicating that financial behaviors, such as digital adoption and investment decisions, often exhibit nonlinear interactions influenced by multi-layered socio-demographic, psychological, and behavioral factors [35, 36]. The modest performance of Random Forest and XGBoost, where precision and recall improved but remained well below acceptable thresholds, suggests that while richer models can begin to decode patterns of engagement, the overlap in behavioral signals between high and low adoption groups presents a persistent challenge.

These results partially confirm the central hypothesis of this study, that machine learning can uncover probabilistic signals of technology acceptance, but the prediction task is inherently difficult due to the class imbalance, limited proxy variables, and lack of temporal context. For example, while features like Customer Satisfaction Score, LTV, and Loyalty Points Earned were found to be influential, their distributions often overlapped significantly across both classes, undermining classification certainty. Moreover, the binary operationalization of technology acceptance, although grounded in behavioral frequency, likely fails to capture the full spectrum of user attitudes, intentions, and constraints that influence digital platform engagement.

The findings further highlight the need for advanced feature engineering and modeling strategies. Given the limited recall achieved by all models for the high acceptance class, future research should consider incorporating time-series features (e.g., user activity over time), contextual metadata (e.g., promotional campaigns or app interface changes), or external behavioral drivers (e.g., economic conditions or peer influence). Additionally, addressing class imbalance through oversampling techniques such as SMOTE, or by applying cost-sensitive learning, may enable models to place greater emphasis on minority class predictions.

From a practical standpoint, the study's insights are relevant for FinTech product managers and developers. The finding that satisfaction and loyalty-related metrics are among the most influential predictors suggests that improving resolution times, reward systems, and user experience design could be effective levers to increase daily usage and long-term adoption of robo-advisory services. Furthermore, the difficulty encountered by even advanced models in classifying high-adoption users reinforces the importance of multi-modal personalization strategies, which go beyond transactional patterns to include psychographic profiling, dynamic behavioral cues, and interactive feedback loops.

In comparison with previous literature, this study advances the field by employing a behaviorally-grounded and data-driven approach to technology acceptance, contrasting with most earlier works that rely heavily on perceptual survey data and structural equation modeling [37, 38]. By focusing on actual user engagement metrics, this research contributes a fresh methodological perspective that complements existing theoretical models like TAM and UTAUT with empirical predictive insights.

Despite these contributions, the study is not without limitations. The use of a single dataset limits generalizability, and the absence of temporal and psychological variables constrains the depth of behavioral modeling. Additionally, while machine learning offers powerful tools for classification, it does not inherently address causality, and interpretability remains a challenge, particularly in black-box models like XGBoost.

The observed discrepancy, where features like LTV and satisfaction show high importance, yet models exhibit low recall, mirrors theoretical expectations from TAM and UTAUT. These models posit that while performance expectancy and facilitating conditions are important, actual adoption is mediated by intention and moderated by personal and environmental variables. Our findings suggest that while behavioral indicators provide partial signals, they are insufficient without accounting for the user's internal decision-making process or surrounding ecosystem. Therefore, weak recall may not indicate model failure, but rather the complexity of behavioral manifestation that lacks visible differentiation in transactional data. To further clarify how our dataset operationalizes theoretical constructs from TAM and UTAUT, Table 2 provides a comparative summary of key constructs, their proxy variables, empirical results, and corresponding interpretations.

In conclusion, this study underscores the potential of machine learning in enhancing our understanding of FinTech user behavior, but also reveals critical areas for methodological refinement. The ability to predict technology acceptance behaviorally is possible, but only to a limited extent, when relying solely on transactional and static user data. To move toward more reliable and actionable models, future research should integrate longitudinal behavioral data, hybrid modeling techniques, and cross-disciplinary insights from behavioral economics, human–computer interaction, and digital psychology.

This study contributes to theoretical discourse by offering an empirical, data-driven lens through which to examine and potentially refine constructs from TAM and UTAUT. While these models traditionally rely on perceptual or intention-based inputs, our ML-based behavioral approach reveals the extent to which such constructs manifest in real-world engagement patterns. The moderate alignment of satisfaction and LTV with perceived usefulness and facilitating conditions demonstrates the potential of behavioral modeling to supplement traditional frameworks. Moreover, the study

Table 2 Comparative table: theoretical constructs, data representation, and empirical insights

Construct (from TAM/UTAUT)	Represented in dataset	Proxy variable used	Empirical finding	Interpretation and implication
Perceived usefulness	Yes	Customer satisfaction score	High feature importance	Aligns with TAM; satisfaction reflects system value contributing to frequent usage
Performance expectancy	Yes	Lifetime value (LTV)	High feature importance	Suggests users adopt when longterm financial benefit is evident
Facilitating conditions	Partially	Issue resolution time, support tickets	Moderate importance	Indicates support access influences adoption, but not strongly predictive alone
Hedonic motivation	Partially	Loyalty points, referral count	Moderate to low importance	These engagement incentives influence behavior but do not fully differentiate high-use users
Social influence	No	Not available	Not tested	Missing from dataset; future studies should include peer or community influence variables
Trust and competence	Indirect	Satisfaction score (proxy), LTV	Suggested through importance of satisfaction	Implied link between good experience and trust, but not directly measured
Digital literacy	No	Not available	Not tested	Important perceptual driver in literature; absence limits generalizability
Behavioral intention	No	Not available	Not applicable	Study focused on actual behavior, not intention. Supports shift toward behavior-centric models

underscores the importance of transitioning from intention-centric models to outcome-oriented behavioral analysis, especially as granular usage data becomes increasingly available in FinTech platforms.

Conclusion

This study set out to explore whether machine learning algorithms can effectively predict behavioral technology acceptance among FinTech users, using real-world transactional and engagement data. By framing daily app usage as a proxy for high technology acceptance, and applying four different classifiers—Logistic Regression, SVM, Random Forest, and XGBoost—the research aimed to uncover the predictive power of user behavior patterns in the adoption of robo-advisory platforms. The results offer partial validation of this objective. While ensemble methods like Random Forest and XGBoost demonstrated better discriminatory performance compared to linear models, their overall ability to detect high-tech-acceptance users was limited, as evidenced by low recall and modest AUC scores. These findings reveal that while behavioral features contain signals of engagement, the distinction between high and low adopters is subtle and non-trivial to model.

While the inclusion of SMOTE improved classification of high-engagement users, challenges remain. This reinforces that behavioral data alone may not be sufficient for robust prediction without temporal depth or psychographic inputs. Our findings are promising but exploratory, and further validation is necessary before real-world deployment.

The study provides several important practical implications for FinTech companies and digital product managers. Firstly, features related to customer satisfaction, loyalty programs, and spending patterns emerged as significant predictors, underscoring the value of customer experience design in promoting sustained app engagement. Companies aiming to increase robo-advisor adoption should consider investing in features that personalize user journeys, reward engagement, and resolve issues efficiently. Secondly, the difficulty encountered by machine learning models in reliably identifying high-engagement users points to the need for real-time behavioral tracking, psychographic profiling, and adaptive interfaces that respond dynamically to user patterns. Incorporating these elements could enhance user trust, retention, and ultimately, technology acceptance.

Despite its contributions, this study has several limitations that must be acknowledged:

- *Single source and cross-sectional dataset*: The analysis relies on one dataset from a single platform, which limits the generalizability of findings across regions, demographics, and types of FinTech services.
- *Behavioral proxy constraints*: While daily app usage is a practical measure of engagement, it may not fully capture deeper cognitive and emotional dimensions of technology acceptance such as trust, perceived usefulness, or risk aversion.
- *Use of only static features*: The model applies static snapshot data without incorporating temporal or sequential behavioral patterns. These patterns are often critical for capturing evolving user behavior and engagement trends.
- *Missing psychographic variables*: Factors such as financial goals, personality traits, or digital literacy levels, which often moderate technology adoption, are not present in the dataset. This limits interpretability.
- *Class imbalance issue*: The relatively small proportion of high-acceptance users posed challenges for predictive performance, especially in recall metrics, despite applying techniques like SMOTE.
- *Lack of causal explanation*: The machine learning models used in this study focus on pattern detection and do not support causal inference. As a result, it is difficult to determine the underlying mechanisms that drive user adoption.

These limitations highlight the importance of future research using longitudinal data, psychographic profiling, temporal behavior logs, and hybrid modeling frameworks that combine machine learning with theoretical or rule-based systems. Such multimodal approaches can improve both predictive performance and theoretical understanding of FinTech adoption.

These limitations open several promising avenues for future research. First, future studies should consider longitudinal data that tracks user interactions over time, enabling more robust modeling of behavioral change and technology internalization. Second, incorporating hybrid modeling frameworks that combine machine learning with rule-based or fuzzy logic systems could enhance prediction in ambiguous or overlapping user groups. Third, addressing class imbalance using advanced techniques such as ensemble bagging, cost-sensitive learning, or generative oversampling (e.g., SMOTE, ADASYN) could improve minority class detection. Lastly, future work should incorporate qualitative variables, such as digital literacy, perceived risk, and financial goals, which may help contextualize engagement patterns in more meaningful ways.

In conclusion, this research offers a novel, data-driven perspective on technology acceptance in FinTech, bridging the gap between behavioral modeling and predictive analytics. While machine learning shows promise in identifying patterns of engagement, its limitations remind us that technology adoption remains a complex human phenomenon, influenced by not only usage metrics but also psychological, social, and contextual factors. Understanding and embracing this complexity will be key to building more inclusive, intelligent, and user-centered financial technologies in the future.

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Data Availability The data will be provided on request.

Declarations

Conflict of interest All authors certify that they have no conflict of interest.

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